

Notes: Baker and Wurgler (JF, 2002)
Market Timing and Capital Structure

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1 Big Picture

Baker and Wurgler ask whether equity market timing is merely a short-run financing phenomenon or whether it has long-run consequences for capital structure. The empirical answer is that the effects are extremely persistent: firms that raised external finance when their market valuations were high subsequently have lower leverage, while firms that raised external finance when valuations were low subsequently have higher leverage. The paper therefore proposes a simple market-timing theory of capital structure: observed leverage is the cumulative historical outcome of past attempts to time the equity market, rather than rapid movement toward a stable target leverage ratio.

Core claim. Capital structure is strongly negatively related to a firm’s historical market valuations, especially valuations prevailing when the firm raised external finance. The finding survives controls for current market-to-book, asset tangibility, profitability, firm size, alternative leverage measures, alternative controls, initial leverage, fixed effects, outliers, and Tobit specifications.

Why the result matters. Standard capital-structure theories generally predict that temporary valuation shocks should not have permanent effects on leverage. Trade-off models imply that firms rebalance toward a target, and pecking-order models imply leverage movements governed by financing deficits and internal funds. Baker and Wurgler instead show that the path of past valuations has explanatory power even a decade later.

2 Research Question and Mechanism

The paper begins from a well-known empirical fact: firms tend to issue equity when market values are high relative to book values and past values, and tend to repurchase when market values are low. The paper’s question is not whether market timing occurs in individual financing decisions, but whether the financing decisions leave a durable imprint on leverage.

Mechanism.

1. Market valuations fluctuate over time, and managers perceive high market-to-book ratios as relatively cheap equity financing opportunities.
2. Firms raise equity when valuations are high, mechanically reducing leverage.
3. If firms do not rapidly undo this reduction through later debt issuance or repurchases, the effect accumulates.
4. Current leverage becomes a historical record of when the firm accessed external capital.

The key empirical variable is the **external-finance-weighted average market-to-book ratio**, denoted here as M/B_{efwa} . It averages a firm’s past market-to-book ratios, weighting years by the amount of external finance raised in those years. High values mean that the firm raised substantial external finance during high valuation periods.

3 Data, Variables, and Empirical Design

Main sample. The authors start with COMPUSTAT firms between 1968 and 1999 and focus heavily on firms for which an IPO date can be identified. The IPO-date sample allows them to track leverage from a clear initial financing event and to construct years relative to IPO.

Leverage.

- Book leverage is book debt divided by total assets.
- Market leverage is book debt divided by assets minus book equity plus market equity.

Financing components. Changes in book leverage are decomposed into parts due to net equity issues, newly retained earnings, and residual growth in assets. This decomposition is crucial because the market-timing hypothesis predicts that valuation effects should operate mainly through equity issuance.

Control variables. The main regressions follow traditional capital-structure controls: current market-to-book, fixed asset intensity, profitability, and firm size. Later robustness tests add Fama-French-style controls, fixed effects, retained earnings, and alternative sample restrictions.

4 Main Findings

- High current market-to-book ratios lower leverage mainly through net equity issues.
- Historical market-to-book ratios, weighted by external finance, have much stronger explanatory power than current market-to-book for older firms.
- The effect of historical valuations remains statistically and economically large after controlling for current market-to-book and other determinants.
- The effect is highly persistent: a large fraction of the initial market-timing effect remains even 10 years later.
- Robustness tests support the same conclusion under book leverage, market leverage, IPO-time samples, all-COMPUSTAT samples, fixed effects, retained earnings controls, endpoint valuation controls, outlier inclusion, and Tobit estimation.

5 Incremental Contribution

Increment relative to prior market-timing evidence. Before this paper, the literature had already documented that firms issue equity when valuations are high, that equity issuers subsequently earn low returns, and that managers say market prices matter for issuance decisions. Baker and Wurgler’s contribution is to trace the cumulative effect of those timing decisions into the cross section of capital structure.

Methodological contribution. The external-finance-weighted average market-to-book ratio is a concise way to summarize a firm’s valuation history at the moments when financing choices mattered most. This distinguishes valuation paths that coincide with financing from valuation paths that do not lead to financing actions.

Conceptual contribution. The paper reframes leverage as path-dependent. Rather than assuming firms have a target debt ratio that they rapidly restore, it proposes that leverage can be a historical residue of past issuance opportunities.

Empirical contribution. The paper provides three independent persistence tests: controlling for current market-to-book, explaining cumulative changes from pre-IPO leverage, and showing that lagged valuation history predicts leverage even 10 years later.

6 Extracted Tables/Figure with Interpretations

Table I. Summary Statistics of Capital Structure and Financing Decisions

Table I
Summary Statistics of Capital Structure and Financing Decisions

Means and standard deviations of leverage and components of the change in assets. Book value leverage is book debt to assets and is expressed in percentage terms. In this and following tables, we drop firm-year observations where book leverage is above one. Market value leverage is book debt divided by the result of total assets minus book equity plus market equity and is expressed in percentage terms. Net equity issues (e/A) are defined as the change in book equity, minus the change in retained earnings, divided by assets. Newly retained earnings ($\Delta RE/A$) are defined as the change in retained earnings divided by assets. Net debt issues (d/A) are defined as the residual change in assets divided by assets. TI sample includes firm-year observations that satisfy data availability, nonfinancial industry, asset, and market-to-book screens. Panel A shows data in years relative to the IPO for firms with a known IPO date. Panel B shows data in calendar time for all COMPUSTAT firms.

Year	N	Book Leverage D/A_t %		Market Leverage D/A_t %		d/A_t %		e/A_t %		$\Delta RE/A_t$ %	
		Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Panel A: IPO Time											
Pre-IPO	2,281	66.54	(21.33)								
IPO	2,839	43.17	(21.41)	27.18	(20.48)	0.35	(20.89)	32.98	(23.50)	9.17	(19.0)
IPO + 1	2,652	45.04	(21.80)	32.78	(23.24)	10.94	(15.97)	6.64	(14.56)	2.68	(13.9)
IPO + 3	2,412	46.88	(21.44)	40.30	(26.15)	5.78	(16.22)	3.93	(13.02)	1.03	(14.9)
IPO + 5	1,668	47.00	(20.99)	43.82	(26.59)	3.53	(15.33)	3.07	(11.28)	1.10	(14.5)
IPO + 10	715	48.68	(20.20)	45.73	(24.85)	4.32	(15.37)	2.44	(9.28)	1.21	(13.6)
Panel B: Calendar Time, All Firms											
1970–1974	4,399	47.22	(17.92)	51.11	(22.79)	4.84	(11.91)	0.98	(4.67)	3.34	(6.5)
1975–1979	5,526	49.11	(17.49)	54.90	(22.27)	4.58	(12.38)	0.66	(3.79)	4.09	(6.7)
1980–1984	5,368	49.71	(19.01)	50.21	(23.84)	3.45	(13.54)	1.70	(6.77)	2.78	(9.0)
1985–1989	6,441	49.44	(20.96)	45.27	(24.31)	3.97	(17.51)	2.11	(9.68)	1.22	(12.3)
1990–1994	7,977	45.98	(20.89)	43.49	(24.69)	2.18	(15.40)	2.54	(10.43)	0.73	(13.5)
1995–1999	11,165	49.01	(21.24)	40.22	(24.35)	4.54	(20.21)	3.54	(13.43)	0.32	(15.7)

Interpretation. The IPO sharply changes capital structure. Book leverage falls from 66.54% pre-IPO to 43.17% at the IPO, and market leverage is especially low at the IPO because high market equity values mechanically lower market leverage. The IPO year is dominated by net equity issuance: e/A averages 32.98%, much larger than debt issuance. After the IPO, average financing shifts toward a steadier mix of debt, equity, and retained earnings. In calendar time, market leverage falls in the late 1990s while equity issuance rises, consistent with high valuations encouraging equity finance.

Table IIa. Determinants of Annual Changes in Leverage: Regression Setup and Panel A

Determinants of Annual Changes in Leverage and Components

OLS regressions of changes in book leverage and its components on the market-to-book ratio, fixed assets, profitability, firm size, and lagged leverage.

$$\left(\frac{D}{A}\right)_t - \left(\frac{D}{A}\right)_{t-1} = a - \left(\frac{S}{A_t}\right) - \left(\frac{\Delta RE}{A_t}\right)_{t-1} - \left[E_{t-1} \left(\frac{1}{A_t} - \frac{1}{A_{t-1}}\right)\right] + a + b \left(\frac{M}{B}\right)_{t-1} + c \left(\frac{PPE}{A}\right)_{t-1} + d \left(\frac{EBITDA}{A}\right)_{t-1} + e \log(S)_{t-1} + f \left(\frac{D}{A}\right)_{t-1} + u_t.$$

We do not report a and f . Book value leverage is book debt to assets and is expressed in percentage terms. The market-to-book ratio is assets minus book equity plus market equity all divided by assets and is measured at time $t-1$ except for the IPO year, when it is measured at time t . We drop firm-year observations where market-to-book is above 10. Fixed assets intensity is defined as net property, plant, and equipment divided by assets. Profitability is defined as operating income before depreciation, divided by assets. Firm size is defined as the log of net sales. The total change in leverage is in Panel A. The net equity issues component is in Panel B. The newly retained earnings component is in Panel C. The growth in assets component is in Panel D. Robust t -statistics are in parentheses.

		M/B_{t-1}		PPE/A_{t-1} %		$EBITDA/A_{t-1}$ %		$\log(S)_{t-1}$		
Year	N	b	t(b)	c	t(c)	d	t(d)	e	t(e)	R ²
Panel A: Change in Book Leverage ($\Delta(D/A_t)$ %)										
IPO	2,281	-3.70	(-11.81)	0.04	(2.50)	-0.10	(-3.84)	3.83	(14.65)	0.45
IPO + 1	2,652	-1.21	(-5.65)	0.04	(3.78)	-0.16	(-6.33)	0.22	(1.24)	0.12
IPO + 3	2,412	-0.93	(-4.47)	0.03	(2.79)	-0.10	(-4.26)	0.69	(4.31)	0.08
IPO + 5	1,668	-0.03	(-0.10)	0.04	(3.89)	-0.11	(-3.22)	0.89	(5.09)	0.06

Interpretation. Panel A shows the net effect of market-to-book on annual leverage changes. The market-to-book coefficient is negative in most IPO-time rows, especially at the IPO and in early post-IPO years. This means that firms with high valuations tend to experience larger decreases in book leverage. Tangibility generally raises leverage, profitability lowers it, and size tends to raise it. The annual-change evidence establishes the short-run mechanical connection between valuation and leverage before the paper turns to persistence.

Table IIb. Determinants of Annual Changes in Leverage: Financing Components

Panel B: Change in Book Leverage Due to Net Equity Issues $(-\epsilon/A_t) \%$							
IPO	2,281	-5.33	(-13.58)	0.05	(2.55)	0.10	(2.89)
IPO + 1	2,652	-3.44	(-11.76)	-0.02	(-1.55)	0.31	(8.82)
IPO + 3	2,412	-3.69	(-10.87)	-0.01	(-1.05)	0.25	(6.26)
IPO + 5	1,668	-3.14	(-6.49)	0.01	(0.54)	0.20	(3.68)
IPO + 10	715	-3.56	(-4.41)	-0.02	(-1.22)	0.25	(2.46)
Panel C: Change in Book Leverage Due to Newly Retained Earnings $(-\Delta RE/A_t)$							
IPO	2,281	0.20	(0.59)	0.02	(1.71)	-0.27	(-7.14)
IPO + 1	2,652	-0.18	(-0.80)	0.01	(0.63)	-0.59	(-16.19)
IPO + 3	2,412	-0.20	(-0.54)	0.03	(2.23)	-0.54	(-12.24)
IPO + 5	1,668	-0.51	(-0.77)	0.01	(0.89)	-0.45	(-5.83)
IPO + 10	715	-0.37	(-0.41)	0.05	(2.44)	-0.56	(-5.20)
Panel D: Change in Book Leverage Due to Growth in Assets $(-E_{t-1}(1/A_t - 1/A_{t-1}))$							
IPO	2,281	1.44	(6.68)	-0.02	(-2.91)	0.07	(3.24)
IPO + 1	2,652	2.41	(9.33)	0.05	(4.70)	0.12	(3.31)
IPO + 3	2,412	2.84	(9.82)	0.01	(0.69)	0.18	(5.59)
IPO + 5	1,668	2.70	(6.50)	0.00	(0.34)	0.25	(4.78)
IPO + 10	715	1.77	(3.84)	0.01	(0.30)	0.29	(3.63)

Interpretation. Panels B-D decompose leverage changes. The strongest valuation effect appears in Panel B: high market-to-book is associated with larger net equity issuance, which reduces leverage. Panel C shows that market-to-book is not consistently related to newly retained earnings, so valuation is not simply forecasting future internal funds. Panel D shows high market-to-book is also associated with asset growth, but the dominant leverage-reducing channel is equity issuance. This is the paper's main channel evidence.

Figure 1. Univariate Explanatory Power of Determinants of Capital Structure

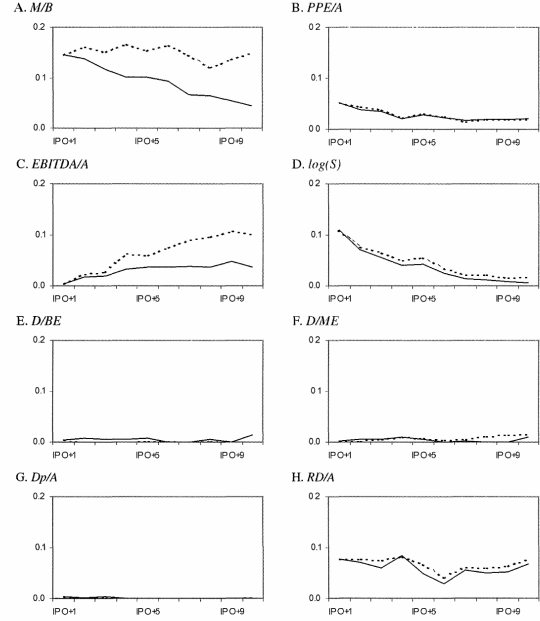


Figure 1. Univariate explanatory power of determinants of capital structure as corporations age. R^2 for univariate OLS regressions of book leverage on determinants of capital structure:

$$\left(\frac{D}{A}\right)_{IPO+t} = a + bX_{t-1} + u_t$$

Book value leverage is book debt to assets. We consider eight independent variables, and each one is defined in two ways. The solid line uses the year $t - 1$ value. The dashed line uses an external finance weighted-average value from the IPO year through year $t - 1$. External finance is defined as net equity issues plus net debt issues. Where this is negative, the weight is set to zero. Market-to-book is assets minus book equity plus market equity all divided by assets. We drop firm-year observations where market-to-book is above 10. Fixed assets intensity is defined as net property, plant, and equipment divided by assets. Profitability is defined as operating income before depreciation divided by assets. Firm size is defined as the log of net sales. Common dividends are scaled by book equity and market equity. Depreciation expense is scaled by assets. R&D is defined as research and development expense divided by assets.

Interpretation. The figure compares the cross-sectional R^2 of current determinants against external-finance-weighted historical determinants. Panel A is the most important: current market-to-book loses explanatory power as firms age, while the weighted historical market-to-book remains powerful. This visual result motivates the later multivariate tests. The gap between the dashed and solid lines says that for older firms, leverage reflects past financing valuations more than the current valuation level. Profitability also accumulates explanatory power, while tangibility, dividends, depreciation, and R&D have weaker or less persistent univariate roles.

Table III. Determinants of Leverage

Determinants of Leverage
OLS and Pama-MacBeth regressions of book and market leverage on the market-to-book ratio, fixed assets, profitability, and firm size.

$$\left(\frac{D}{A}\right)_t = a + b\left(\frac{M}{B}\right)_{t-1} + c\left(\frac{M}{B}\right)_{t-2} + d\left(\frac{PPE}{A}\right)_{t-1} + e\left(\frac{EBITDA}{A}\right)_{t-1} + f \log(S)_{t-1} + u_t.$$

We do not report a . Leverage is defined either as book debt to book assets (book value) or book debt to the result of total assets minus book equity plus market equity (market value) and is expressed in percentage terms. The market-to-book ratio is defined in two ways. The first is a weighted average market-to-book ratio from the IPO year to year $t-1$. The weights are the amount of external finance raised in each year. External finance is defined as net equity issues plus net debt issues. Where this is negative, the weight is set to zero. The second is the market-to-book ratio in year $t-1$, defined as assets minus book equity plus market equity all divided by assets. Fixed assets intensity is defined as net property, plant, and equipment divided by assets. Profitability is defined as operating income before depreciation divided by assets. Firm size is defined as the log of net sales. Panel A shows results for book leverage. Panel B shows results for market leverage. Robust t -statistics are in parentheses.

Year	N	b	$t(b)$	c	$t(c)$	d	$t(d)$	e	$t(e)$	f	$t(f)$	R^2
Panel A: Book Leverage %												
IPO + 1	2,632			-4.36	(-15.59)	0.13	(7.30)	-0.22	(-6.44)	5.00	(16.40)	0.25
IPO + 3	2,412	-4.93	(-8.40)	-0.86	(-1.90)	0.12	(6.83)	-0.31	(-7.41)	4.62	(15.53)	0.25
IPO + 5	1,668	-6.49	(-9.78)	0.05	(0.07)	0.12	(5.74)	-0.32	(-7.18)	4.30	(12.40)	0.26
IPO + 10	715	-10.81	(-10.89)	3.71	(3.32)	0.12	(3.65)	-0.38	(-5.01)	2.67	(4.82)	0.23
1980-1999 All firms	31,151	-7.21	(-31.20)	2.20	(3.38)	0.04	(0.62)	-0.48	(-7.20)	2.84	(21.79)	0.20
Panel B: Market Leverage %												
IPO + 1	2,694			-8.09	(-26.57)	0.14	(8.00)	-0.19	(-4.26)	2.91	(9.96)	0.36
IPO + 3	2,482	-6.05	(-10.03)	-5.84	(-9.83)	0.12	(6.50)	-0.32	(-9.17)	3.31	(10.43)	0.40
IPO + 5	1,731	-7.41	(-8.55)	-5.30	(-6.14)	0.12	(4.99)	-0.36	(-7.09)	2.43	(6.25)	0.37
IPO + 10	738	-10.77	(-9.98)	-3.29	(-3.19)	0.11	(3.22)	-0.53	(-5.92)	1.23	(2.06)	0.37

Interpretation. This is the central leverage-level regression. The coefficient on M/B_{efwa} is negative and highly significant for both book and market leverage, even when current market-to-book and the standard Rajan-Zingales controls are included. The key interpretation is that historical valuations at financing dates explain leverage beyond current investment opportunities or contemporaneous valuation effects. The result becomes particularly striking at IPO+10, where the weighted historical market-to-book remains strongly negative.

Table V. Determinants of Cumulative Changes in Leverage from the Pre-IPO Value

Determinants of Cumulative Changes in Leverage from the Pre-IPO Value
OLS and Pama-MacBeth regressions of the cumulative change in leverage since the pre-IPO value on the market-to-book ratio, fixed assets, profitability, and firm size.

$$\left(\frac{D}{A}\right)_t - \left(\frac{D}{A}\right)_{pre-IPO} = a + b\left(\frac{M}{B}\right)_{t-1} + c\left(\frac{M}{B}\right)_{t-2} + d\left(\frac{PPE}{A}\right)_{t-1} + e\left(\frac{EBITDA}{A}\right)_{t-1} + f \log(S)_{t-1} + g\left(\frac{D}{A}\right)_{t-1} + u_t.$$

We do not report a and g . Leverage is defined either as book debt to book assets (book value) or book debt to the result of total assets minus book equity plus market equity (market value) and is expressed in percentage terms. The market-to-book ratio is defined in two ways. The first is a weighted average market-to-book ratio from the IPO year to year $t-1$. The weights are the amount of external finance raised in each year. External finance is defined as net equity issues plus net debt issues. Where this is negative, the weight is set to zero. The second is the market-to-book ratio in year $t-1$, defined as assets minus book equity plus market equity all divided by assets. Fixed assets intensity is defined as net property, plant, and equipment divided by assets. Profitability is defined as operating income before depreciation divided by assets. Firm size is defined as the log of net sales. Panel A shows results for book leverage. Panel B shows results for market leverage. Robust t -statistics are in parentheses.

Year	N	$M/B_{book,t-1}$	$M/B_{mkt,t-1}$	PPE/A_{t-1}	$EBITDA/A_{t-1}$	$\log(S)_{t-1}$						
		b	c	$t(c)$	d	$t(d)$	e	$t(e)$	f	$t(f)$	R^2	
Panel A: Book Leverage %												
IPO + 1	2,166		-4.10	(-13.43)	0.11	(6.95)	-0.27	(-6.52)	3.49	(12.44)	0.46	
IPO + 3	1,553	-4.45	(-6.48)	-0.75	(-1.94)	0.11	(5.34)	-0.36	(-6.25)	3.45	(9.70)	0.46
IPO + 5	1,058	-5.46	(-6.73)	0.03	(0.03)	0.11	(4.33)	-0.36	(-7.35)	3.58	(8.34)	0.47
IPO + 10	381	-10.26	(-6.96)	4.25	(3.39)	0.07	(1.82)	-0.28	(-3.54)	2.34	(3.26)	0.45
1980-1999 All firms	17,283	-6.81	(-16.72)	2.54	(5.03)	0.05	(7.19)	-0.50	(-12.33)	2.61	(14.54)	0.48
Panel B: Market Leverage %												
IPO + 1	2,180		-8.10	(-21.93)	0.12	(6.98)	-0.27	(-7.33)	1.57	(5.26)	0.54	
IPO + 3	1,588	-5.47	(-7.70)	-5.68	(-7.57)	0.12	(5.36)	-0.40	(-8.96)	2.14	(5.41)	0.53
IPO + 5	1,091	-6.14	(-6.54)	-5.14	(-5.03)	0.12	(4.18)	-0.39	(-5.86)	1.59	(3.23)	0.49
IPO + 10	393	-11.78	(-8.29)	-1.96	(-1.76)	0.09	(2.10)	-0.47	(-6.17)	0.69	(0.91)	0.56

Interpretation. Table V shifts the dependent variable from leverage levels to cumulative changes from the pre-IPO level. This is an important design change because it removes the possibility that M/B_{efwa} merely proxies for a firm's initial leverage type. The results remain negative and significant, meaning that historical valuation episodes actually move firms away from their initial leverage positions. This strengthens the path-dependence interpretation.

Table IV. Determinants of Leverage: Alternative Control Variables

Year	N	M/B_{book}	M/B	ET/A %	Div/BE %	$Div/ME\%$	Dp/A %	RD/A %	RDD	$\log(A)$	R^2
	b	$t(b)$	c	$t(c)$	d	$t(d)$	e	$t(e)$	f	$t(f)$	R^2
Panel A: Book Leverage %											
IPO + 1	2,651		-4.75	(-16.83)	-0.20	(6.92)	-0.54	(-5.19)	0.06	(0.78)	0.28
IPO + 3	2,411	-4.46	(-7.86)	-0.36	(-0.66)	-0.35	(-0.99)	-0.16	(-0.39)	-0.08	0.29
IPO + 5	1,667	-6.13	(-9.02)	0.25	(0.35)	-0.30	(-0.78)	-0.14	(-0.37)	-0.32	0.26
IPO + 10	715	-9.45	(-8.52)	3.84	(3.46)	-0.44	(-5.51)	0.44	(2.90)	-0.17	0.27
1980-1999 All firms	31,147	-6.28	(-28.59)	1.88	(3.69)	-0.43	(-7.99)	0.19	(2.55)	-0.48	0.22
Panel B: Market Leverage %											
IPO + 1	2,693		-7.66	(-26.11)	-0.22	(-7.09)	-0.04	(-1.42)	0.13	(1.75)	0.39
IPO + 3	2,481	-5.43	(-8.95)	-4.95	(-8.05)	-0.36	(-1.18)	0.42	(3.28)	-0.30	0.44
IPO + 5	1,730	-6.53	(-8.51)	-5.22	(-5.51)	-0.36	(-1.08)	0.22	(2.67)	-0.32	0.40
IPO + 10	738	-8.17	(-6.50)	-3.10	(-2.91)	-0.51	(-8.83)	0.07	(0.70)	-0.60	0.40
1980-1999 All firms	32,205	-5.89	(-20.76)	-5.41	(-12.02)	-0.53	(-9.50)	0.17	(2.86)	-0.38	0.39

Interpretation. Table IV asks whether the historical valuation effect is actually standing in for other persistent firm characteristics. Adding alternative controls such as dividends, depreciation, R&D intensity, an R&D dummy, and log assets does not remove the negative M/B_{efwa} coefficient. This supports the view that the market-timing result is not a disguised profitability, payout, investment-opportunity, or non-debt-tax-shield effect.

Table VI. Persistence of Market-to-Book Effects

Persistence of Market-to-Book Effects
Pama-MacBeth regressions of current and future leverage on the market-to-book ratio, fixed assets, profitability, and firm size.

$$\begin{aligned} \left(\frac{D}{A}\right)_{t+1} &= a_1 + b_1\left(\frac{M}{B}\right)_{t,market} + c_1\left(\frac{M}{B}\right)_{t,firm} + d_1\left(\frac{PPE}{A}\right)_{t-1} + e_1\left(\frac{EBITDA}{A}\right)_{t-1} + f_1 \log(S)_{t-1} + u_{1,t+1} \\ \left(\frac{D}{A}\right)_{t+3} &= a_3 + b_3\left(\frac{M}{B}\right)_{t,market} + c_3\left(\frac{M}{B}\right)_{t,firm} + d_3\left(\frac{PPE}{A}\right)_{t-1} + e_3\left(\frac{EBITDA}{A}\right)_{t-1} + f_3 \log(S)_{t-1} + u_{3,t+3} \\ \left(\frac{D}{A}\right)_{t+5} &= a_5 + b_5\left(\frac{M}{B}\right)_{t,market} + c_5\left(\frac{M}{B}\right)_{t,firm} + d_5\left(\frac{PPE}{A}\right)_{t-1} + e_5\left(\frac{EBITDA}{A}\right)_{t-1} + f_5 \log(S)_{t-1} + u_{5,t+5} \end{aligned}$$

We report only b_1 , b_3 , b_5 , and c_5 . Leverage is defined either as book debt to book assets (book value) or book debt to the result of total assets minus book equity plus market equity (market value) and is expressed in percentage terms. The persistence of the market-to-book effect is measured by a ratio of coefficients: b_3 divided by b_1 or b_5 divided by b_1 . For each τ from 1 to 10, we run a set of three regressions for each year t starting τ years prior to 1980 and ending τ years prior to 1999 and record 30 sets of these coefficients. Only firms that survive τ years are included so that each set of coefficients is calculated with the same sample of firms. A lower bound estimate (LB) for the ratio of the coefficients is calculated for the five percent level of significance by drawing from the joint distribution of the two means. Panel A shows results for book leverage. Panel B shows results for market leverage. Robust t -statistics are in parentheses.

Year	b_1	$t(b_1)$	b_3	$t(b_3)$	b_5	$t(b_5)$	c_5	$t(c_5)$	b_3/b_1	$LB_{b3/b1}$	b_5/b_1	$LB_{b5/b1}$
Panel A: Book Leverage %												
$t+1$	-7.21	(-21.20)	-7.21	(-21.20)	-7.21	(-21.20)	2.30	(3.38)	1.00	—	1.00	—
$t+3$	-7.96	(-20.39)	-8.31	(-18.44)	-5.87	(-15.33)	0.30	(0.29)	0.86	0.81	0.76	0.71
$t+5$	-7.48	(-24.85)	-5.89	(-14.98)	-5.35	(-15.97)	0.24	(0.29)	0.80	0.73	0.73	0.67
$t+10$	-6.91	(-13.77)	-5.08	(-12.13)	-5.11	(-15.15)	0.45	(0.49)	0.73	0.66	0.75	0.68
Panel B: Market Leverage %												
$t+1$	-7.25	(-20.32)	-7.25	(-20.32)	-7.25	(-20.32)	5.53	(-14.45)	1.00	—	1.00	—
$t+3$	-7.50	(-19.34)	-7.51	(-18.33)	-6.44	(-16.64)	-7.72	(-19.33)	1.00	0.90	0.90	0.81
$t+5$	-7.65	(-19.04)	-7.61	(-17.57)	-6.13	(-16.26)	-8.57	(-16.92)	1.00	0.89	0.82	0.75

Interpretation. Table VI directly studies whether the effect decays. It compares the original coefficient with coefficients predicting future leverage up to 10 years later. For book leverage, about 73% of the initial effect remains after 10 years; for market leverage, the effect is essentially permanent by the paper's estimates. The lower-bound calculations also remain large. This table is the strongest challenge to fast target-adjustment views of capital structure.

Table A1. Robustness Tests for Leverage and Cumulative Leverage Changes

Table A1
Determinants of Leverage and Cumulative Changes in Leverage from the Pre-IPO Level: Robustness Tests

OLS and Fama–MacBeth regressions of leverage and the cumulative change in leverage from the pre-IPO level on the market-to-book ratio, fixed assets, profitability, and firm size. The leverage regressions use equation (5) as a model. The leverage changes regressions use equation (6) as a model. Other independent variables are included as indicated. We report only the coefficient on the weighted average market-to-book. Leverage is defined as book debt to book assets (book value) or book debt to the result of total assets minus book equity plus market equity (market value) and is expressed in percentage terms. The weighted average market-to-book ratio is the average market-to-book ratio between the IPO year to year $t - 1$, weighted by the amount of external finance raised in each year. External finance is defined as net equity issues plus net debt issues. Where this is negative, the weight is set to zero. Three types of robustness tests are described in the Appendix. The first set (SIC-3 fixed effects, IPO-year fixed effects, retained earnings) add additional controls. The retained earnings variable is the total retained earnings as of t , scaled by assets. The second set (start $M/B = \text{end } M/B$, outliers included) uses different initial samples. We consider firms that at IPO + 10 had the same market-to-book (within 0.5 in absolute value) as at the IPO. We also consider firms with market-to-book outliers. The third set (Tobit) uses an alternative estimation approach. Panels A (book value) and C (market value) use firms that had IPOs 10 years earlier. Panels B (book value) and D (market value) use all firms with COMPUSTAT data between 1980 and 1999. Robust t -statistics are in parentheses.

	Leverage Level				Leverage Change from Pre-IPO			
	N	b	$t(b)$	R^2	N	b	$t(b)$	R^2
Panel A: Book Leverage %, IPO + 10								
SIC-3 fixed effects	715	-8.55	(-6.45)	0.45	381	-10.54	(-4.80)	0.69
IPO-year fixed effects	715	-10.82	(-9.59)	0.26	381	-9.77	(-6.10)	0.48
Retained earnings	703	-10.53	(-9.80)	0.32	379	-9.78	(-6.32)	0.53
Start $M/B = \text{End } M/B$	299	-21.26	(-5.83)	0.31	168	-24.48	(-6.01)	0.52
Outliers included	716	-10.58	(-10.39)	0.23	381	-10.26	(-6.96)	0.45
Tobit	715	-10.81	(-10.29)	—	381	-10.26	(-7.06)	—
Panel B: Book Leverage %, 1980–1999 All Firms								
SIC-3 fixed effects	31,151	-6.46	(-17.69)	0.39	17,383	-6.10	(-15.46)	0.64
IPO-year fixed effects	31,151	-7.43	(-21.11)	0.22	17,383	-6.49	(-16.23)	0.50
Retained earnings	30,762	-6.43	(-15.71)	0.33	17,272	-6.30	(-14.80)	0.54
Start $M/B = \text{End } M/B$	17,149	-8.51	(-11.19)	0.18	9,822	-9.08	(-8.60)	0.46
Outliers included	31,243	-6.93	(-18.11)	0.20	17,428	-6.49	(-15.41)	0.48
Tobit	31,151	-7.21	(-21.20)	—	17,383	-6.81	(-16.72)	—
Panel C: Market Leverage %, IPO + 10								
SIC-3 fixed effects	738	-7.78	(-5.19)	0.55	393	-11.24	(-5.20)	0.73
IPO-year fixed effects	738	-10.51	(-8.76)	0.43	393	-11.06	(-6.75)	0.59
Retained earnings	726	-10.35	(-8.29)	0.40	391	-10.70	(-6.82)	0.57
Start $M/B = \text{End } M/B$	312	-17.95	(-3.58)	0.39	175	-21.88	(-5.42)	0.56
Outliers included	739	-10.10	(-8.52)	0.35	393	-11.78	(-8.29)	0.56
Tobit	738	-10.77	(-9.38)	—	393	-11.78	(-7.83)	—
Panel D: Market Leverage %, 1980–1999 All Firms								
SIC-3 fixed effects	32,209	-6.10	(-17.45)	0.51	17,896	-5.82	(-14.18)	0.69
IPO-year fixed effects	32,209	-7.71	(-19.71)	0.37	17,896	-6.59	(-14.70)	0.57
Retained earnings	31,803	-5.73	(-19.46)	0.42	17,779	-5.73	(-12.03)	0.58
Start $M/B = \text{End } M/B$	17,679	-5.77	(-6.87)	0.33	10,098	-5.97	(-4.40)	0.55
Outliers included	32,313	-7.87	(-13.99)	0.34	17,945	-6.48	(-14.75)	0.54
Tobit	32,209	-7.35	(-20.52)	—	17,896	-6.74	(-14.81)	—

Interpretation. The appendix table repeats the key regressions under alternative specifications. SIC-3 fixed effects and IPO-year fixed effects address industry and cohort concerns; retained earnings controls address the possibility that the effect is really accumulated profitability; endpoint valuation restrictions ask whether the path matters even when beginning and ending market-to-book are similar; outlier inclusion checks sample trimming; and Tobit accounts for bounded leverage. The M/B_{efwa} coefficient remains negative and statistically strong across these tests, reinforcing the robustness of the market-timing interpretation.

7 How to Read the Evidence Together

The paper’s empirical logic is cumulative. Table II first shows that market-to-book affects annual leverage changes through equity issuance. Figure 1 then shows that historical market-to-book has increasing relative explanatory power as firms age. Table III turns this into a multivariate regression result, while Table IV rules out several alternative controls. Table V shows that valuation history explains changes away from initial capital structure. Table VI shows that the effect persists for a decade. Table A1 checks that the pattern is not a fragile artifact of industry, cohort, retained earnings, outliers, or bounded dependent variables.

8 Relation to Theories of Capital Structure

Trade-off theory. A standard trade-off model can explain why current market-to-book is correlated with leverage if market-to-book proxies for investment opportunities, risk, agency costs, or tax effects. It has difficulty explaining why valuation shocks from a decade earlier still predict leverage after current variables are controlled for.

Pecking order. The pecking order predicts reluctance to issue equity because of adverse selection. A dynamic version can explain precautionary low leverage before investment needs, but it does not naturally predict that firms with high past valuations should remain persistently less levered.

Managerial entrenchment. Entrenchment theories can generate a failure to rebalance after equity issuance, but Baker and Wurgler emphasize that the evidence also fits managers’ attempts to exploit new investors during high-valuation periods.

Market timing. The simplest interpretation is that managers issue equity when equity seems cheap and then do not undo the leverage effect. Capital structure becomes the accumulated outcome of these historical financing choices.

9 Critical Assessment

The paper is influential because it turns a behavioral or opportunistic financing idea into a capital-structure theory. Its strongest feature is the persistence evidence: the result is not only that high valuation firms issue equity today, but that the consequences remain visible many years later. The strongest empirical concern is interpretation. Market-to-book can capture mispricing, growth opportunities, risk, or omitted firm traits. The paper addresses this concern through controls and robustness tests, but the data cannot fully distinguish rational time-varying adverse selection from irrational mispricing. The authors acknowledge this and present market timing as the most natural explanation rather than the only possible one.

10 Takeaways

1. Market timing affects capital structure not just immediately, but persistently.
2. The relevant valuation variable is not simply current market-to-book, but valuation at the times the firm raised external capital.
3. Equity issuance is the key channel through which high valuations reduce leverage.
4. Historical valuation paths matter even after controlling for current valuation, tangible assets,

profitability, size, and alternative controls.

5. The paper challenges target-leverage models by showing that past temporary valuation shocks can have near-permanent leverage effects.